

3.2.1 What are the feeding relationships between living organisms?

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 1			
Radiation from the Sun is the source of energy for living organisms.		4.4.1.1	4.2.2.5
Green plants and algae absorb a small amount of the light that reaches them and make glucose by photosynthesis. These organisms are called producers. Carbon dioxide + water → glucose + oxygen Students should know the word equation for photosynthesis.	Suggested activity for TDA Investigate the rate of photosynthesis in pond weed.	4.4.1.1	4.2.2.5
Outcome 2			
Animals and plants may be adapted for survival in the conditions where they normally live.	Students should be able to explain how organisms are adapted to live in their natural environment given appropriate information in image or prose format. Examples may include polar bears in the Arctic, or cacti in deserts. Suggested activity for TDA Investigate the use of choice chambers, eg with woodlice or maggots.	4.7.1.4	
Outcome 3			
Feeding relationships within a community can be represented by a food chain. All food chains begin with a producer. A food web can be used to understand the interdependence of species within an ecosystem in terms of food resources.		4.7.2.1	4.4.2.1
Outcome 4			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
All materials in the living world are recycled to provide the building blocks for future organisms.		4.7.2.2	4.4.1.2
Decay of dead plants and animals by microorganisms returns carbon to the atmosphere as carbon dioxide to be used by plants in photosynthesis.	The names of particular microorganisms are not required. Suggested activity for TDA Investigate the change in temperature as grass cuttings decay.	4.7.2.2	4.4.1.2